# Daniel Mora

New York, NY · Open to relocation within the US

**AI Systems & Research Engineer**

damorasoler@gmail.com · daniel-mora.com · github.com/daMS2005 · huggingface.co/dams2005 · LinkedIn

## PROFILE

I build machine-learning systems end to end, from point-in-time data pipelines and self-supervised pretraining to custom GPU kernels and unattended agentic pipelines. My research uses sparse autoencoders to test whether language models and the human brain rely on the same features.

## EDUCATION

### IE University

Madrid, Spain · 2026

Bachelor in Computer Science and Artificial Intelligence

## SELECTED WORK

### JEPA Transformer — ArgusJEPA + CassaFormer

PyTorch · Triton · 2026

Two-stage forecaster for US equities; code on GitHub, weights released on Hugging Face

- Built a point-in-time data platform (IBKR, SEC filings by acceptance time, FRED/ALFRED vintages, scored news) on PostgreSQL + DuckDB over a survivorship-free top-500 universe.
- Stage I: self-supervised JEPA encoder predicting an EMA teacher's 256-d embedding from a 50%-masked view; frozen as daily embeddings.
- Stage II: 2.05M-parameter causal decoder (RoPE, QK-RMSNorm, top-1 MoE, Student-t head), trained separately for auditable leakage boundaries.
- Fused Triton kernels, kept only where they beat `torch.compile` on an L4: MoE experts 1.2–4.1× per layer, 1.1–1.5× end to end.

### TC_ACC — Autonomous Documentary Studio

Python · LangGraph · Remotion · 2026

Agentic production system; full code private, public excerpt on GitHub

- A case reference in; a researched, narrated, scored and rendered 10-minute documentary out in about five hours, with no human intervention.
- 29 stages, 6 departments, 2 independent review gates; typed Pydantic artifacts, resumable stages, 46 CI tests passing with all provider keys blanked.

### Brain–LLM Feature Correspondence — Bachelor thesis

Supervisor: Dr. Luciano Dyballa

Draft paper (ICLR format): "A Cross-Description Test of Sparse Feature Correspondence Between Language Models and fMRI Responses"

- Tested whether brain and LM mappings rely on the same sparse features: 59 participants, 200 cortical parcels, six SAE systems (1B–12B).
- Built SAE feature discovery, encoding models and structured temporal nulls; the calibrated result bounds strong correspondence instead of overclaiming.

### Sparse-autoencoder releases

PyTorch · Hugging Face · GCP

- **OLMo SAE Builder:** single-GPU pipeline that streamed ∼20M token activations per layer from OLMo-3-7B-Think and released seven SAELens-compatible TopK SAEs.
- **Triplet SAE Project:** compared SAE feature steering with LoRA for factual triplet extraction; shipped a fine-tuned Gemma 3 4B with 4-/8-bit GPTQ exports.

## EXPERIENCE

### Freelance web developer

Client work

Solo client builds: SENSE Travel Agency, Reusa (React, Next.js, Tailwind, Vercel), Dr. Evangelina Soler medical practice

### Team software projects — IE University

- **Cloud Banking Platform:** one of two full-stack developers (Vue, Flask, Bicep on Azure, GitHub Actions CI/CD).
- **LearnMate:** built the agent flows, backend and first Streamlit prototypes for an agentic study partner (team of seven).

### Front-end Developer Intern — MAI-Book

Remote · Jun–Oct 2023

Earlier: IE Start-Up Lab (GPAX, Jan–Jul 2023) and IE Corporate Relations internship (Jun 2023)

## LEADERSHIP

**Founder & President, IE Dominican Republic Club** — grew its flagship event into the university's largest student-led diversity event (500+ attendees). **Business Development Officer, IE Investment Club** — secured sponsorships with financial-sector firms.

## SKILLS

| | |
|---|---|
| **ML & research** | PyTorch, Triton, Hugging Face (Transformers, PEFT/LoRA, SAELens), sparse autoencoders, JEPA, MoE, encoding models |
| **Systems** | Python, C++, SQL, TypeScript; LangGraph, Pydantic, FastAPI; PostgreSQL, DuckDB, Redis; Docker, GitHub Actions, GCP, Azure |
| **Languages** | Spanish (native), English (fluent), French (B1) |